

INTEGRATED DIGITAL HOUSE RENTAL AND TAX MANAGEMENT SYSTEM

RHTMS

A prototype platform connecting property owners, tenants, housing offices and tax offices across the residential rental lifecycle.

Software Project Management — Final Presentation

Background and Motivation

- Residential rental registration and taxation in Addis Ababa involve multiple offices: a housing administration bureau and a separate tax/revenue process.
- During our field investigation, we found the Addis Ababa Housing Development and Administration Bureau already operates a digital portal ("RHRCAS" / Mesob One-Stop-Shop) for landlord, tenant, house, and contract records, observed directly at two housing offices during our visit.
- However, based on our interviews, tax-side verification of a rental agreement still depends on the physical, printed agreement being presented at the tax office — the two offices are not connected by a shared digital record.
- Legal grounding referenced in this project: Housing Rent Control and Management Proclamation No. 1320/2016; Addis Ababa Directive No. 184/2025 (Residential Rent Control and Management); Income Tax (Amendment) Proclamation No. 1395/2025.
- **This motivated RHTMS: a prototype that models the full lifecycle — registration, housing approval, tenant registration, agreement, tax assessment, and payment — as one connected, role-based system.**

Problem Statement

- Based on our interviews and observations, we identified the following problems:
 - The existing housing portal and the tax office are not connected — a physical agreement must be carried between offices for tax verification.
 - No single, role-based workflow connects property owner, tenant, housing officer, and tax officer around one shared record.
 - Verifying a rental agreement's authenticity outside an office visit is difficult — there is no independent, self-service check.
 - Rental-income tax calculation is easy to get wrong when the exact rule isn't clearly specified — our own early prototype briefly used an incorrect flat rate before it was corrected against a real published source (see Slide 16).
- *Scope note: these findings come from a limited set of stakeholder interviews and site observations, not a comprehensive survey of every housing or tax office in Ethiopia.*

Existing Process and Identified Gaps

Existing step (observed)	What we found	Gap our project addresses
Housing/contract registration	A digital portal (RHRCAS) already records landlords, tenants, houses, and contracts	Not connected to the tax office's own process
Tax verification	The physical, printed agreement is presented at the tax office	No digital handoff or shared record between housing and tax
Agreement authenticity check	Done visually / manually by office staff	No independent, self-service verification (e.g., QR code)
Cross-office status visibility	Not observed during our visits	No shared dashboard across owner, tenant, housing, and tax roles
Rental-income tax calculation	Handled by tax office staff	Risk of an incorrectly specified or applied rate without a structured, sourced model

Important: this does not mean the existing process is purely paper-based — a real digital housing portal exists (see Slide 6). The gap is the missing connection between housing records and tax verification.

Stakeholder Investigation and Data Collection

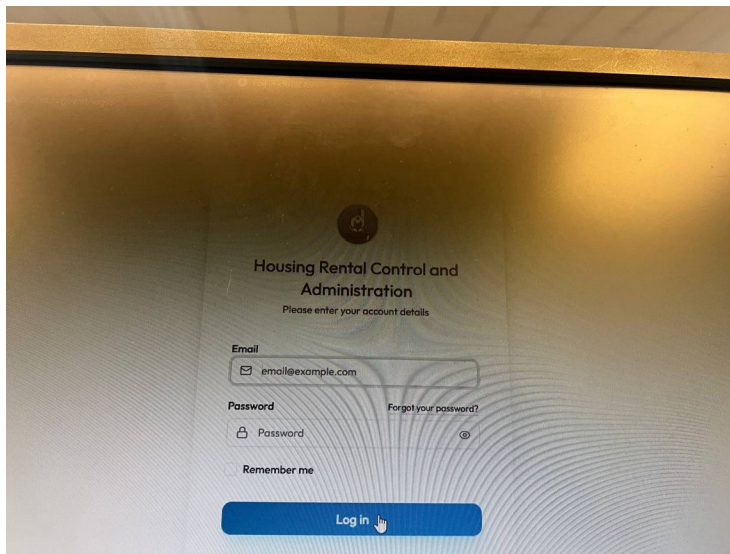
Stakeholder group	Who we spoke with
Tenant	Osman Muhammad
Property Owner	Solomon Hailu
Housing Officers	Getalem (Kirkos sub-city) · Meqdes (Lemi Kura sub-city)
Tax Officers	Yimam Yimer · Dula (Arada sub-city)

- **Purpose of the interviews:**
 - Understand the existing rental registration and agreement process.
 - Identify housing approval activities and tax document verification steps.
 - Find manual, disconnected, or repetitive steps.
 - Understand how tenants/owners interact with offices.

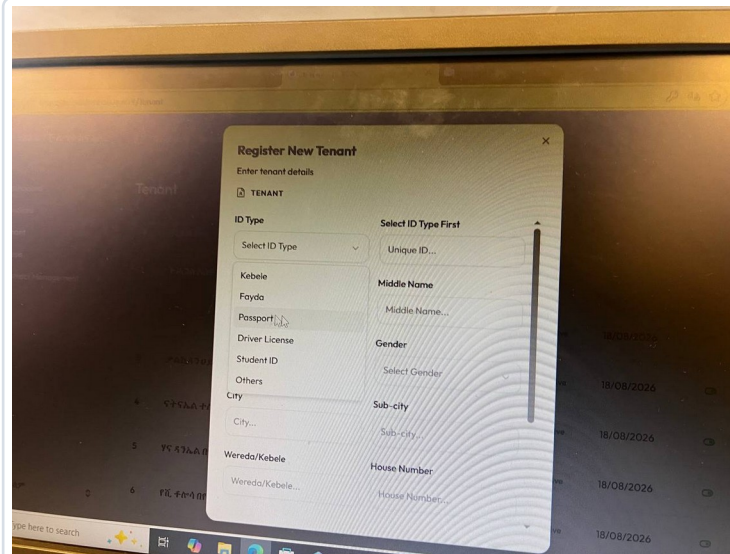
- **Main findings (based on our interviews and observations):**
 - A digital housing portal already exists and is used at the housing offices we visited (Kirkos, Lemi Kura).
 - Tax offices (Arada) still rely on the physical agreement being brought in for verification and assessment.
 - There is no shared digital record linking housing-side registration to tax-side assessment.
- **These findings directly shaped our requirements: QR-based public verification, role-based dashboards connecting housing and tax officers, and a structured, sourced tax-bracket assessment tied to the agreement record.**

These interviews and observations are a limited stakeholder sample used to inform this prototype's design — not a claim of representing every tenant, owner, or office in Ethiopia.

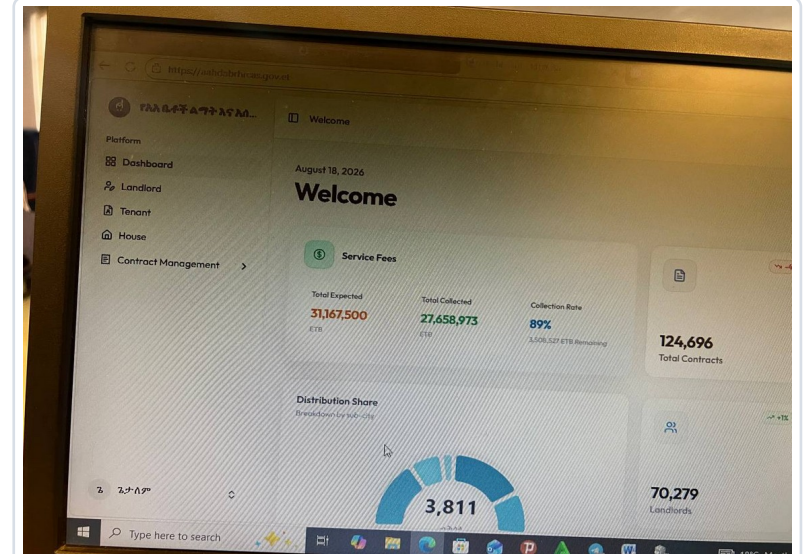
Existing System: A Digital Portal Already In Use



The housing bureau's existing portal login screen, observed during our site visit.



Its "Register New Tenant" form — structured data entry already exists for tenant records.

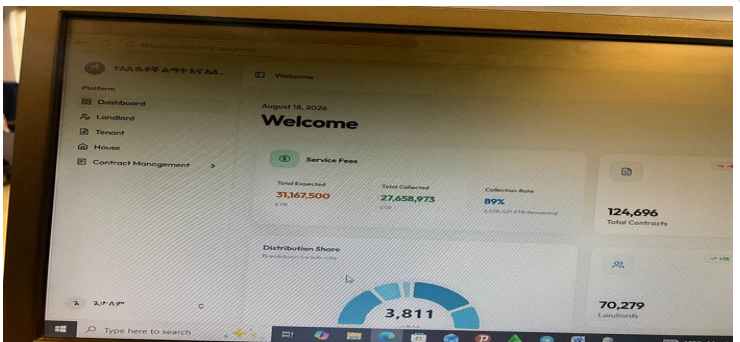


Its live dashboard: 124,696 total contracts, 70,279 landlords, 89% service-fee collection rate.

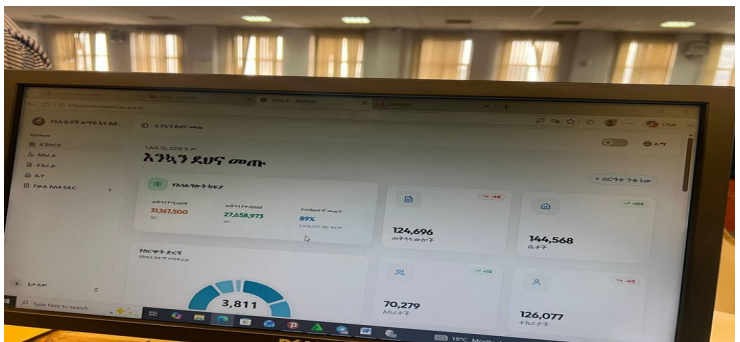
- This confirms the existing process is not purely manual: a real, operational government digital portal already handles landlord, tenant, house, and contract records at real scale.
- **Identified limitation (from our interviews): this portal's records are not connected to the tax office's own verification and assessment step — that gap is what RHTMS's prototype targets.**

Multilingual Support and the Physical Agreement Handoff

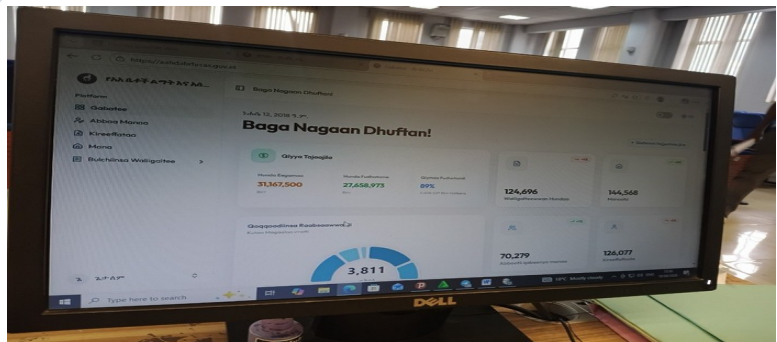
The same existing portal already serves English, Amharic, and Afaan Oromoo:



English

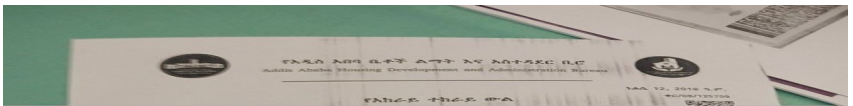


Amharic (አማርኛ)



Afaan Oromoo

This directly echoes RHTMS's own English / Amharic / Afaan Oromoo interface (Slide 17).



[Tenant / owner names, phone numbers, address, rent amount and signature/stamp redacted for privacy]

Document structure preserved:
Parties / Tenant / Leased Property / Term / Conditions

A rental agreement issued by the Addis Ababa Housing Development and Administration Bureau. Personal details (names, phone numbers, address, rent amount, signature/stamp) are redacted for privacy; only the document header/template is shown.

- This physical agreement is prepared and presented during the existing administrative and tax-related process — consistent with what our tenant, owner, and officer interviews described.
- This is observed evidence, not a claim that RHTMS has legally replaced this physical document or any government system.
- **It motivated two concrete features: QR-based public contract verification, and a shared, structured tax-assessment record tied directly to the digital agreement.**

Proposed Solution and Project Objectives

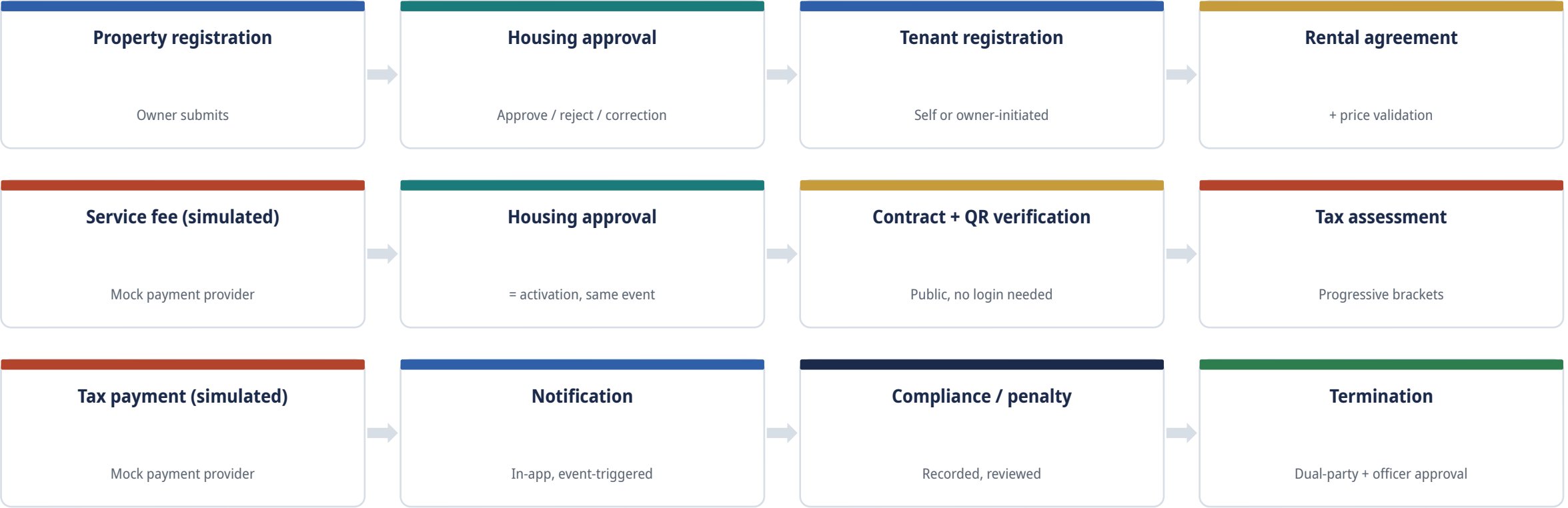
- **Proposed solution**
 - One shared, role-based, database-backed prototype connecting property owners, tenants, housing officers, and tax officers around the same agreement record — from registration through tax assessment.
- **General objective**
 - Design and implement a digital platform modeling the residential rental lifecycle, connecting the roles identified in our stakeholder investigation under role-based access.
- **Specific objectives**
 - Property registration, housing review/approval, tenant registration, and rental agreements with price validation.
 - A verifiable digital contract (QR code) and a sourced, progressive rental-income tax assessment.
 - Role-based access control with record-level ownership checks and a full audit trail.
 - An interface in English, Amharic, and Afaan Oromoo.

Scope, Stakeholders, and System Roles

Property Owner Registers properties, creates rental agreements, pays service fees, views tax status	Tenant Is registered onto agreements, pays rent-related fees, requests termination	Housing Officer Reviews/approves properties and agreements, reviews penalties
Tax Officer Assesses rental-income tax against the active bracket table	Super Admin Manages users, tax brackets, penalty/price rules, audit log	Public / Verifier Scans a contract's QR code to confirm it is genuine — no login required

In scope: the roles and workflow above, as a prototype. Out of scope: direct integration with the government RHRCAS portal or any real payment/banking network (see Limitations, Slide 19).

Main System Workflow (Implemented)



Every payment step is a simulated/mock provider (CBE, Telebirr, Bank Transfer, Cash) — no real transaction occurs. All other steps shown are implemented against the real database.

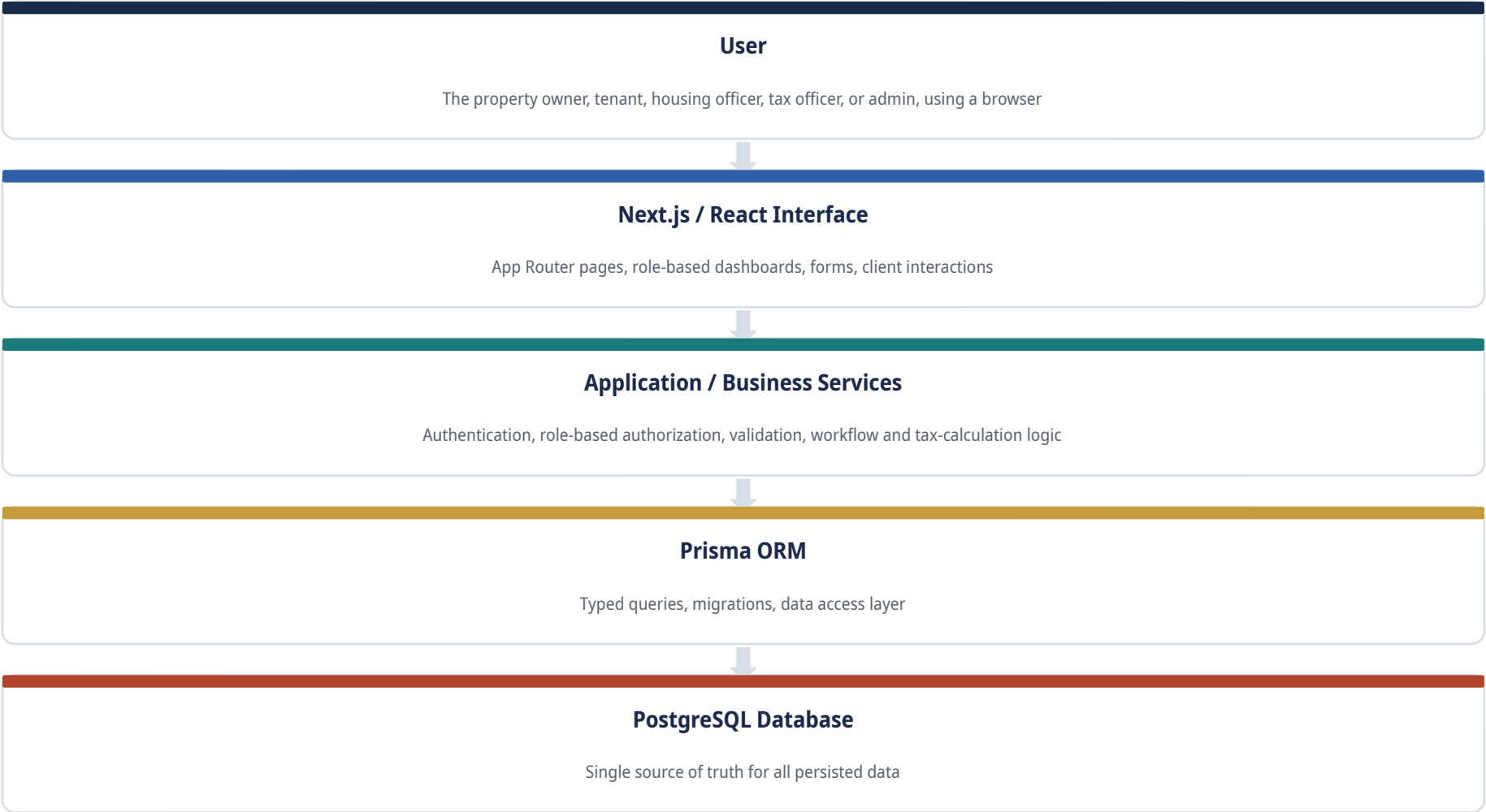
Functional Requirements

Requirement	Status	Evidence
User authentication	Implemented	Auth.js credentials + JWT sessions
Role-based access control	Implemented	Central permission map + middleware role gate
Property registration & approval	Implemented	Owner submit → officer approve/reject/correction
Tenant registration	Implemented	Self-service or owner-initiated
Rental agreement + price validation	Implemented	Configured expected price ranges
Rental price-increase regulation	Implemented	2-year wait, 11.5% cap — see Slide 16
Tax assessment	Implemented (prototype)	Progressive brackets, sourced (see Slide 16)
Payment record management	Prototype / Simulated	Mock multi-provider abstraction
QR / public verification	Implemented	Public /verify/[token] page
Notifications	Partially implemented	In-app only; no SMS/email or scheduling
Audit logs	Implemented	Actor, action, before/after value
Reports & admin configuration	Implemented	Live revenue figures, rule management
Real government/bank integration	Not implemented (planned)	See Future Work

Non-Functional Requirements

Category	Requirement	Status
Security	Only authorized users act on records they own or their role covers	Implemented — see Slide 18
Auditability	Every sensitive change is traceable	Implemented — AuditLog
Data integrity	Historical records never silently overwritten	Implemented — versioned agreement history
Usability	Interface usable in the user's preferred language	Implemented — English/Amharic/Afaan Oromoo
Maintainability	Business rules changeable without a code deploy	Implemented — configurable database rules
Reliability	Core logic verified without a browser	Implemented — automated validation scripts
Performance / Scalability	Response time and load under many users	Not formally tested — future validation needed

System Architecture



A layered, modular-monolith architecture — one deployable application, internally separated by responsibility. Not a microservices system.

Database / Major Entities

Identity & Access User, OwnerProfile, TenantProfile, StaffProfile	Property Property, Ownership, PropertyDocument, PropertyReview, SubCity, Woreda
Agreement RentalAgreement, AgreementVersion, AgreementReview, RentalPriceRule	Tax TaxRule, TaxBracket, TaxAssessment
Compliance & Payments PenaltyRule, Penalty, Payment	Admin / Audit Notification, AuditLog, SystemConfiguration

23 Prisma models in total. Key design choice: agreement price changes are versioned (AgreementVersion), never overwritten, and tax rates are stored as editable TaxBracket rows, not hard-coded.

Project Management and Team Collaboration



- **Communication**

- The team communicated through a Telegram group — sharing progress and informing others when work was completed.

- **Version control**

- GitHub was used for code collaboration. The repository uses feature branches (e.g. develop, feat/*) alongside main, with work reviewed before being integrated.

- **Review and change process**

- Issues and changes were discussed before or during implementation; feedback and approval were communicated through the team group and the development workflow itself.
- *No formal Scrum ceremonies (sprints, daily stand-ups, retrospectives) were used — this reflects our actual, lighter-weight process, not an oversight.*

Implementation Status and Major Developed Features

- Property lifecycle — registration, housing review (approve/reject/correction), status-gated agreement creation.
- Agreement lifecycle — creation, price validation, service-fee payment, housing approval → activation, contract/QR issuance, renewal, termination.
- Tax (prototype) — progressive rental-income tax brackets, sourced from PwC Worldwide Tax Summaries — Ethiopia; corrected during development from an earlier, incorrect flat-rate assumption.
- Rental price-increase regulation — a 2-year waiting period and an 11.5% cap on any rent increase, kept strictly separate from the tax calculation above.
- Payments (simulated) — a mock multi-provider abstraction (CBE, Telebirr, Bank Transfer, Cash); no real transaction occurs.
- Notifications — in-app, triggered by real business events (approvals, payments, tax assessment, penalties).
- Localization — English, Amharic, Afaan Oromoo across the implemented workflow.
- Admin console — user management, tax-bracket/penalty/price-rule management, audit log, live revenue reports.

Security, Authentication, and Role-Based Access Control

- Authentication — Auth.js (NextAuth v5), credentials provider with bcrypt-hashed passwords, JWT session strategy.
- Role-based access control — a central permission map defines what each of the 5 roles may do; every server action re-checks it.
- Record-level authorization — ownership checks stop one owner from reading or writing another owner's property or agreement.
- Route-level role enforcement — every dashboard is gated in request middleware so one role cannot open another role's dashboard by navigating directly to its URL.
- *Development lesson: during a final review, a cross-role dashboard access issue was identified — a role-specific page guard was not being reliably re-checked on every request. It was corrected by enforcing the check in request middleware and re-verified across all five roles. This reflects that role-based access requires continuous testing, not a one-time implementation.*
- Audit trail — every write to a sensitive record is logged with actor, action, and before/after values.

PROPERTY_OWNER

TENANT

HOUSING_OFFICER

TAX_OFFICER

SUPER_ADMIN

Testing and Validation

Area	Validation method	Result
Type safety	npm run typecheck (tsc)	Pass
Code quality	npm run lint (ESLint)	Pass
Localization	verify:i18n script — catalog + key parity across 3 locales	Pass — 424 keys matched
Tax calculation	verify:tax — bracket, boundary, zero-rate, invalid-input tests	Pass
Rent-increase rule	verify:rent-increase — waiting period, cap, decrease cases	Pass
End-to-end lifecycle	verify:workflow — 16-step scripted run against the real database	Pass
Role-based access, logout, QR verification	Manual HTTP/session-level testing (curl with real sessions)	Confirmed
Full UI/browser regression, load/performance testing	Not performed in this project	Requires further validation

Limitations and Future Improvements

- **Limitations**

- Payment integrations are simulated — no real bank or mobile-money connection.
- No direct integration with the existing government housing portal (RHRCAS) observed during our investigation.
- Tax brackets are sourced, but a small rounding difference can occur exactly at a bracket boundary.
- The rent-increase figures (2 years, 11.5%) are project-owner-supplied requirements, not independently verified against a cited proclamation article.
- No production deployment; performance/load testing not yet performed.

- **Future Work**

- Explore integration with authorized government systems, where permitted.
- Real payment-provider integration behind the existing interface.
- Expanded stakeholder testing and formal performance/security testing.
- SMS/email notifications, document upload, and a tenant-initiated request-to-rent flow.

CONCLUSION

- Our stakeholder investigation found that Addis Ababa's housing offices already use a digital portal, but tax verification still depends on a physical agreement moving between offices, with no shared record connecting the two.
- RHTMS is our prototype response: a single, role-based platform carrying a property and its agreement from registration through housing approval, tax assessment, and termination.
- Across development, we identified and corrected real defects in our own implementation — a terminology error, a cross-role access-control gap, and an incorrect tax-rate model — and implemented the rental price-increase rule as a feature kept strictly separate from tax.
- **Key contribution: a working, evidence-based prototype — grounded in real stakeholder interviews and a real observed system, with disclosed limitations and a clear boundary between what is implemented, what is simulated, and what remains future work.**

Our current prototype supports a connected, role-based, auditable rental and tax workflow — built and corrected through an evidence-based development process.